Constants (Elementary Charges)

(-)coulomb $= 6.28\times10^{18}$ e's

$k = 9\times10^{9}\ N\cdot m^2/C^2$

$\epsilon_0 = 8.85\times10^{-12}\ C^2/N\cdot m^2$

$c = 3\times10^{8}\ m/s$

$e = 1.6\times10^{-19}\ C$

$m_p = 1.67\times10^{-27}\ kg$

$m_e = 9.11\times10^{-31}\ kg$

$\mu_0 = 4\pi\times10^{-7}\ T\cdot m/A$

$h = 6.63\times10^{-34}\ J/s$

Prefixes:

$P = 10^{15}$	$f = 10^{-15}$
$T = 10^{12}$	$p = 10^{-12}$
$G = 10^{9}$	$n = 10^{-9}$
$M = 10^{6}$	$\mu = 10^{-6}$
$k = 10^{3}$	$m = 10^{-3}$
$h = 10^{2}$	$c = 10^{-2}$

Geometry:

Cylinder

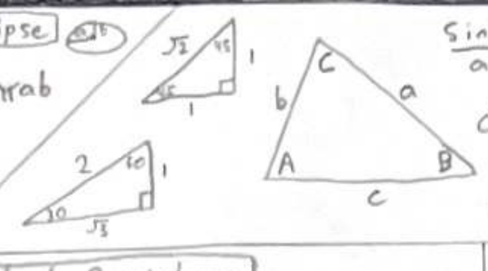
Sphere $A = 2\pi r^2 + 2\pi r h$, $V = \pi r^2 h$

$A = 4\pi r^2$, $V = \frac{4}{3}\pi r^3$

Cone $A = \pi r(r + \sqrt{h^2 + r^2})$, $V = \pi r^2 (\frac{h}{3})$

Circle $C = 2\pi r$, $A = \pi r^2$

Ellipse $A = \pi ab$



$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

$c^2 = a^2 + b^2 - 2ab\cos C$

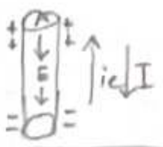
$\sin^2\theta + \cos^2\theta = 1$

$\sin(2\theta) = 2\sin\theta\cos\theta$

27

$N_e = i_e \Delta t$, $i_e = N_e A v_d$

where v_d = drift speed, A = area, i_e = electrons per second

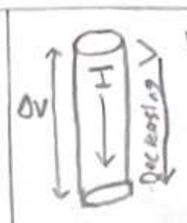


Conventional Current $I = e\cdot i_e$, $I = \frac{Q}{\Delta t}$

Current Density $I = J\cdot A$, $J = \sigma\cdot E$

Conductivity $\sigma = \frac{n_e e^2 \tau}{m}$

Resistivity $\rho = \frac{1}{\sigma} = \frac{m}{n_e e^2 \tau}$



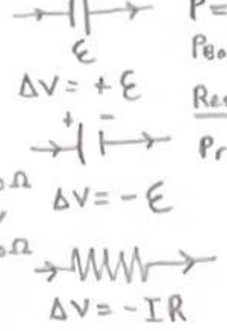
$E = \frac{\Delta V}{L}$

$I = \frac{\Delta V}{R}$

$R = \frac{\rho L}{A}$

28 DC Circuits

Loop Law $\sum_{loop} V_i = 0$



$\Delta V = +\mathcal{E}$

$\Delta V = -\mathcal{E}$

$\Delta V = -IR$

Ohm's Law $I = \frac{\Delta V}{R}$

Junction Law $\sum I_{in} = \sum I_{out}$

$I_1 + I_2 = I_3$

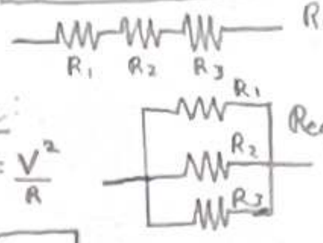
"Capacitors in parallel all have the same ΔV"

"Capacitors in series all have the same charge"

Power $P = IV$, $P_{bat} = \mathcal{E}V$

Resistor dissipate: $P_r = IV = I^2R = \frac{V^2}{R}$

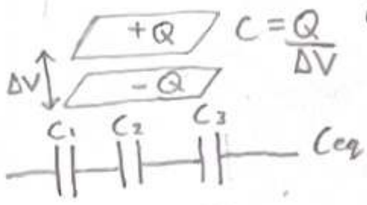
Equivilant Resistance



$R_{eq} = R_1 + R_2 + R_3$

$R_{eq} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)^{-1}$

Capacitors



$C = \frac{Q}{\Delta V}$

$C_{eq} = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)^{-1}$

$C_{eq} = C_1 + C_2 + C_3$

Dieletric

Parallel plate: $C = \frac{\epsilon_0 A k}{d}$, $C = k C_0$ where k is the dielectric constant

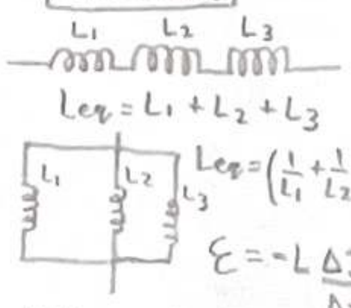
Energy "Capacitors store energy in the electric field"

$U_c = \frac{1}{2}C(\Delta V)^2$

Density

$U_E = \frac{1}{2}k\epsilon_0 E^2$ where k is dielectric constant

Inductors



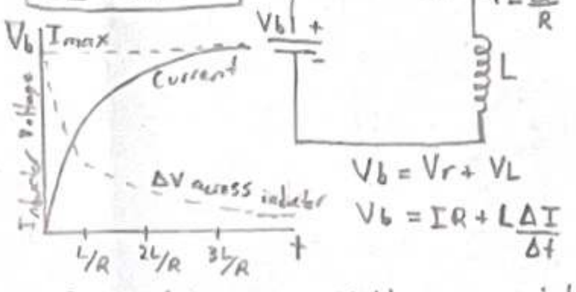
$L_{eq} = L_1 + L_2 + L_3$

$L_{eq} = \left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}\right)^{-1}$

$\mathcal{E} = -L\frac{\Delta I}{\Delta t}$

$\frac{\text{Volt}\cdot\text{second}}{\text{Ampere}} = \text{Henry}$

RL Circuit



$V_b = V_r + V_L$

$V_b = IR + L\frac{\Delta I}{\Delta t}$

Current: $I = \frac{V_b}{R}(1 - e^{-tR/L})$

Voltage across inductor: $V_L = V_b e^{-tR/L}$

When $t=0$: $I = 0$, $V_r = 0$, $V_L = V_b$

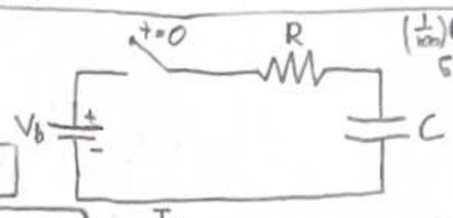
When $t=\infty$: $I = \frac{V_b}{R}$, $V_r = V_b$

*Note $\tau = \frac{L}{R}$

Electric Field

$\vec{E} = \frac{\Delta V}{d}$

RC Circuit



$(\frac{1}{100})RC$ = Short, $5RC$ = Long

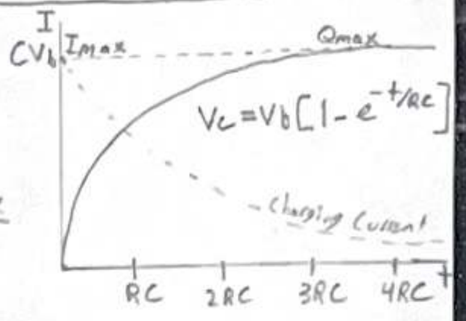
Charging Capacitor

$V_b = V_r + V_c$

$V_b = IR + \frac{Q}{C}$

Charge on Capacitor: $Q = CV_b[1 - e^{-t/RC}]$

Charging Current: $I = \frac{V_b}{R}e^{-t/RC}$



When $t=0$: $Q = 0$, $V_c = 0$, $I = \frac{V_b}{R}$

When $t=\infty$: $Q = CV_b$, $V_c = V_b$, $I = 0$

Discharging Capacitor

$V_c = \frac{Q}{C} = IR$

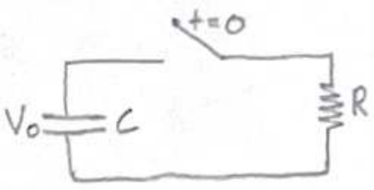
@ $t=0$: $V_c = V_0$, $Q = CV_0$, $I = \frac{V_0}{R}$

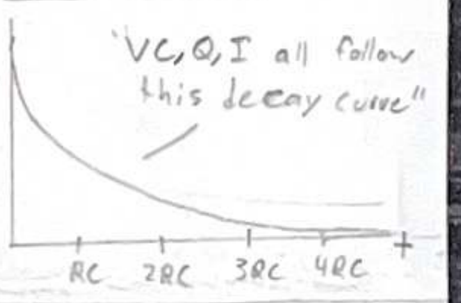
$V_c = V_0 e^{-t/RC}$

$Q = CV_0 e^{-t/RC}$

$I = \frac{V_0}{R}e^{-t/RC}$

*Note $\tau = RC$

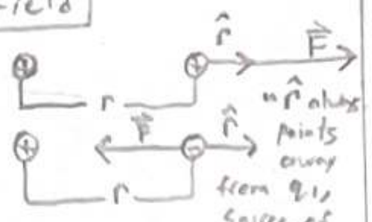




"V_c, Q, I all follow this decay curve"

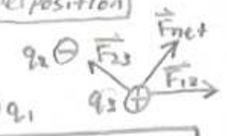
22/23 Electric Field

Coulomb's Law $\vec{F} = \frac{k q_1 q_2}{(K) r^2}\hat{r}$



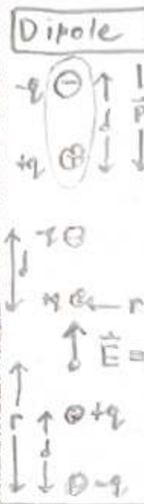
"$\hat{r}$ always points away from q_1, source of the field"

Superposition



Electric Field of a point Charge $\vec{E} = \frac{1}{4\pi\epsilon_0}\cdot\frac{q}{r^2}\hat{r} = \frac{kq}{r^2}\hat{r}$

Dipole Moment $\vec{p} = q\vec{d}$



$\vec{E} = \frac{1}{4\pi\epsilon_0}\cdot\frac{p}{r^3}$

$\vec{E} = \frac{1}{4\pi\epsilon_0}\cdot\frac{2p}{r^3}$

$\vec{E}$ field

$\vec{F} = q\vec{E}$

$\vec{a} = \frac{q\vec{E}}{m}$

Charge Density

Linear: $\lambda = Q/L$ (charge per meter)

Surface: $\sigma = Q/A$ (charge per square meter)

Volume: $\rho = Q/V$ (per meter cubed)

Dipole in an $\vec{E}$ field

$\tau = pE\sin\theta$

24 Gauss's Law

$\oint \vec{E}\cdot d\vec{A} = \frac{q}{\epsilon_0}$

$\Phi_E = \oint \vec{E}\cdot d\vec{A}$

① if $\vec{E} \parallel \vec{A}$; $\Phi_E = 0$

② if $\vec{E} \perp \vec{A}$; $\Phi_E = EA$

gaussian surface

* if $\Phi_E = 0$; there is no enclosed charge and vice versa

Point Charge: $\vec{E} = \frac{kQ}{r^2}$

Sphere of Uniform Charge: $\vec{E} = \frac{kQ}{r^2}$; $r > R$; $\vec{E} = \frac{kQr}{R^3}$; $r < R$

Conducting Cylinder: (∞)

$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 r}$; $r \geq R$

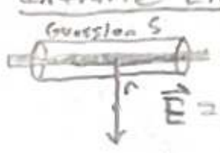
$\vec{E} = 0$; $r < R$

Conducting Sphere:

$\vec{E} = \frac{kQ}{r^2}$, $r > R$

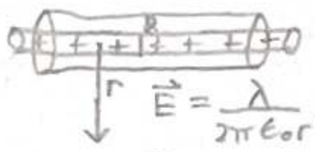
$\vec{E} = 0$; $r < R$

Infinite Line:



$\vec{E} = \frac{\lambda}{2\pi r\epsilon_0}$

Uniform Cylinder of Charge: (∞)



$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 r}$; $r \geq R$

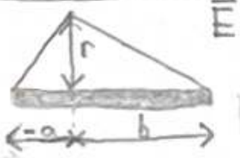
$\vec{E} = \frac{\lambda r}{2\pi\epsilon_0 R^2}$; $r < R$

Sheet of Charge: (∞)

$\vec{E} = \frac{\sigma}{2\epsilon_0}$

Parallel Plate Capacitor: $\vec{E} = \begin{cases} \frac{\sigma}{\epsilon_0}; & \text{inside} \\ 0; & \text{outside} \end{cases}$

Finite Line:



$\vec{E} = \frac{k\lambda}{r}\left[\frac{b}{(r^2+b^2)^{1/2}} + \frac{a}{(r^2+a^2)^{1/2}}\right]$

Ring of Charge:

$\vec{E} = \frac{(k)(\lambda)(2\pi)(R)(r)}{(r^2+R^2)^{3/2}}$

Disk of Charge:

$\vec{E} = k\sigma 2\pi\left[1 - \frac{r}{\sqrt{r^2+R^2}}\right]$

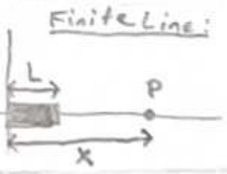
Pipe of Charge:

$\vec{E} = \frac{\rho(R_1^2 - R_2^2)}{2r\epsilon_0}$

Uniform Slab of Charge: (∞)

$\vec{E} = \begin{cases} \frac{\rho d}{\epsilon_0}; & \text{outside} \\ \frac{\rho z}{\epsilon_0}; & \text{inside} \end{cases}$

Finite Line:



$\vec{E} = \frac{k\lambda L}{x(x-L)}$

25 Electric Potential

Point Charge $V = \frac{kQ}{r}$

Superposition $V = \frac{kQ_1}{r_1} + \frac{kQ_2}{r_2} + \frac{kQ_3}{r_3}$

Dipole: $V = \frac{kp\cos\theta}{r^2}$; $r \gg d$

Conservation of Energy $KE_i + qV_i = KE_f + qV_f$

Change in potential $V_a - V_b = kQ\left[\frac{1}{r_a} - \frac{1}{r_b}\right]$

Potential Energy $U = q\cdot V$

Point Charge: (2) $U = \frac{kq_1q_2}{r}$

Dipole: $U = -pE\cos\theta$

Line of Charge

$V = k\lambda\ln\left[\frac{b + \sqrt{b^2+d^2}}{-a + \sqrt{a^2+d^2}}\right]$

Ring of charge

$V = \frac{k\lambda 2\pi R}{E}$

Disk of Charge

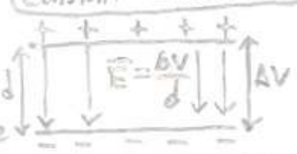
$V = k\sigma 2\pi\left[\sqrt{z^2+R^2} - z\right]$

Conducting Sphere

$V = \frac{kQ}{R}$; inside

$V = \frac{kQ}{r}$; outside

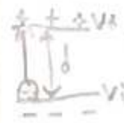
Constant Electric Field



$\vec{E} = \frac{\Delta V}{d}$

Work $W = q\Delta V$

Voltage from $\vec{E}$ field



$V_a - V_i = -Ed$

$V_a - V_i = -Ed\cos\theta$

$V_a - V_i = -\int \vec{E}\cdot d\vec{s}$

$\Delta V = \frac{dW}{q} = \int \vec{E}\cdot d\vec{s}$

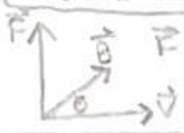
$\vec{E}$ field from Voltage

$\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$

$\vec{E} = -\frac{dV}{dx}\hat{i} - \frac{dV}{dy}\hat{j} - \frac{dV}{dz}\hat{k}$

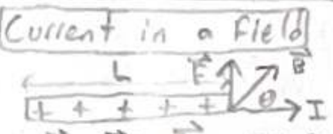
29/30 Magnetic Field

Lorentz Force Law



$\vec{F} = q\vec{v}\times\vec{B} = qvB\sin\theta$

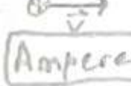
Current in a field



$\vec{F} = I\vec{L}\times\vec{B} = ILB\sin\theta$

Bio-Savart Law

Moving Point Charge:



$\vec{B} = \frac{\mu_0}{4\pi}\cdot\frac{q\vec{v}\times\hat{r}}{r^2}$

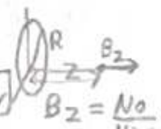
Long Straight Wire

$\vec{B} = \frac{\mu_0 I}{2\pi r}$

Center of a Current Loop

$\vec{B} = \frac{\mu_0 I}{2R}$

On-axis of Loop



$B_z = \frac{\mu_0}{4\pi}\cdot\frac{2\pi R^2 I}{(z^2+R^2)^{3/2}}$

Amperes Law

$\oint \vec{B}\cdot d\vec{r} = \mu_0 I + \mu_0\epsilon_0\frac{d\Phi_E}{dt}$

Inside a long Solenoid

$\vec{B} = \frac{\mu_0 N I}{L}$

Cylinder Conductor

Current density: $I = J\pi R^2$

$B = \frac{\mu_0 r I}{2\pi R^2}$; $r < R$

$B = \frac{\mu_0 I}{2\pi R}$; $r = R$

$B = \frac{\mu_0 I}{2\pi r}$; $r > R$

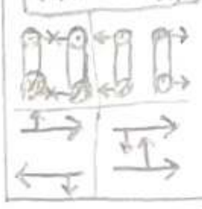
Inside a Toroid

$B = \frac{\mu_0 N I}{2\pi r}$ where r is the centerwidth

Force Between Wires

$F = \frac{\mu_0 I_1 I_2}{2\pi r}\cdot L$

Interactions



Torque Loop $\vec{\tau} = \vec{\mu}\times\vec{B} = \mu B\sin\theta$ where $\vec{\mu} = I\vec{A}$

Cyclotron

radius of orbit: $r_{cyc} = \frac{mv}{qB}$

$f_{cyc} = \frac{qB}{2\pi m}$

Magnetic Flux

$\Phi_M = \vec{A}\cdot\vec{B} = AB\cos\theta$

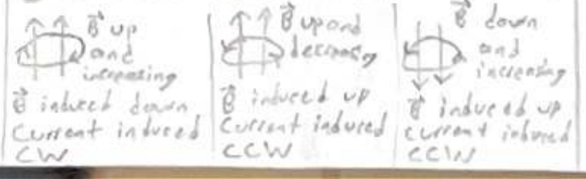
Magnetic Guass Law $\oint B\cdot d\vec{A} = 0$

How to change Flux

① Loop moves through a $\vec{B}$ field

② Loop changes area or rotates

③ $\vec{B}$ field increases or decreases



$\vec{B}$ up and increasing → $\vec{B}$ induced down, current induced CW

$\vec{B}$ up and decreasing → $\vec{B}$ induced up, current induced CCW

$\vec{B}$ down and increasing → $\vec{B}$ induced up, current induced CCW

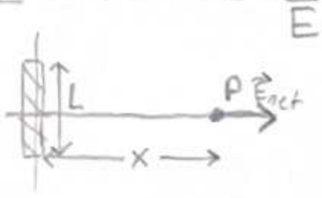
Shere Uniform with hole:

$V_{tot} = \frac{4}{3}\pi(R^3 - a^3)$

$V_i = \frac{4}{3}\pi(r^3 - a^3)$

$\vec{E} = \frac{kQ}{r^2}\left(\frac{r^3 - a^3}{R^3 - a^3}\right)$

Finite Line:



$\vec{E} = \frac{(k)(\lambda)(L)}{x(x-L)}$

Faraday's Law

$\mathcal{E} = \left|\frac{d\Phi_M}{dt}\right|$

Two ways to induce Current:

① Motional EMF — $\mathcal{E} = VLB$, $\Delta V = VLB$

② Changing Magnetic Field — $\oint \vec{E}\cdot d\vec{r} = -\frac{d\Phi_M}{dt}$

Voltage Generated $= \frac{-N\Delta(BA)}{\Delta t}$

Induced Current: $I = \frac{\mathcal{E}}{R}$

Inductors

Solenoid: $L = \frac{\mu_0 N^2 A}{L}$

Potential Difference: $\Delta V_L = -L\frac{\Delta I}{\Delta t}$

Energy stored in $\vec{B}$ Field: $U_L = \frac{1}{2}LI^2$

Energy density: $U_B = \frac{1}{2\mu_0}B^2$

LC Circuit

Charge on the Capacitor: $Q(t) = Q_0\cos\omega t$

Current through the inductor: $I = \omega Q_0 \sin\omega t = I_{max}\sin\omega t$

Angular frequency: $\omega = \sqrt{\frac{1}{LC}}$

LR Circuit "Discharge"

$\Delta V_r + \Delta V_L = 0$

$I_0 = \frac{\Delta V_{bat}}{R}$

$I = I_0 e^{-tR/L}$

32 AC Circuits

Root Mean Squared

$I_{rms} = \frac{I_{max}}{\sqrt{2}}$

$V_{rms} = \frac{V_{max}}{\sqrt{2}}$

$\mathcal{E}_{rms} = \frac{\mathcal{E}_0}{\sqrt{2}}$

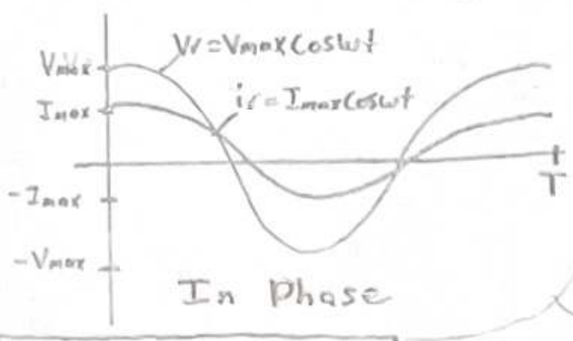
Resistor Circuit

$V_{max} = \mathcal{E}_0$ (For this)

resistor Voltage: $V_r = i_r R$, $V_r = V_{max}\cos\omega t$

Instantaneous Current: $i_r = \frac{V_r}{R} = \frac{V_{max}\cos\omega t}{R}$, $i_r = I_{max}\cos\omega t$

$I_{max} = \frac{V_{max}}{R}$



Capacitor Circuits

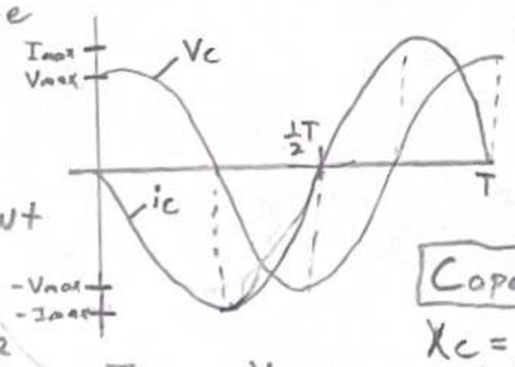
V_{max} is the max voltage across the Capacitor*

$V_{max} = \mathcal{E}_0$ (For this)

$V_c = V_{max}\cos\omega t$

Charge on Capacitor: $q = CV_c = CV_{max}\cos\omega t$

instantaneous Current: $i_c = \omega C V_c \cos\left(\omega t + \frac{\pi}{2}\right)$



Out of Phase: Current leads Voltage by $\pi/2$

Capacitive Reactance

$X_c = \frac{1}{\omega C}$

*Large at Low frequencies

$I_{max} = \frac{V_{max}}{X_c}$ "or" $V_{max} = I_{max}X_c$

Inductor Circuit

$V_{max} = \mathcal{E}_0$ (For this)

$V_L = V_{max}\cos\omega t$

V_{max} is the max voltage across the inductor

instantaneous Current: $i_L = I_{max}\cos\left(\omega t - \frac{\pi}{2}\right)$

where $I_{max} = \frac{V_{max}}{X_L}$

Inductive Reactance: $X_L = \omega L$

$I_{max} = \frac{V_{max}}{X_L}$

*Large at high frequencies

Out of Phase: Voltage leads Current by $\pi/2$

RC Filter Circuit

$\mathcal{E}_0^2 = (V_r)_{max}^2 + (V_c)_{max}^2$

Peak Values

$I_{max} = \frac{\mathcal{E}_0}{\sqrt{R^2 + X_c^2}}$

*Note $(I_r)_{max} = (I_c)_{max} = I_{max}$

$(V_r)_{max} = (I_{max})R = \frac{\mathcal{E}_0 R}{\sqrt{R^2 + X_c^2}}$

$(V_c)_{max} = (I_{max})X_c = \frac{\mathcal{E}_0 X_c}{\sqrt{R^2 + X_c^2}}$

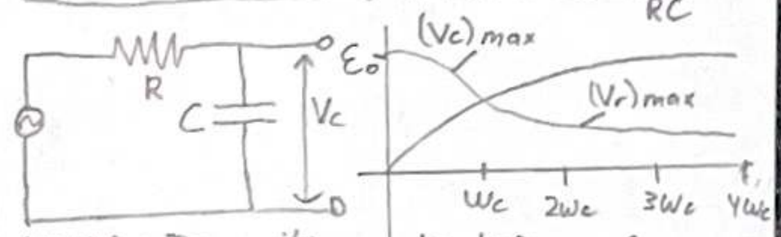
Crossover Frequency

$f_c = \frac{\omega_c}{2\pi}$

ω_c occurs when $(V_r)_{max} = (V_c)_{max}$

$\omega_c = \frac{1}{RC}$

Low Pass Filter



$\omega < \omega_c$; Transmits

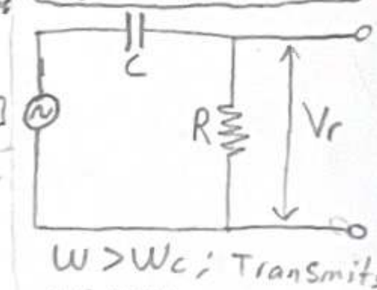
$\omega > \omega_c$; Blocks

Lets Low frequecies pass

$\lim_{\omega\to 0}(V_c)_{max} = \mathcal{E}_0$

$\lim_{\omega\to\infty}(V_r)_{max} = \mathcal{E}_0$

High Pass Filter

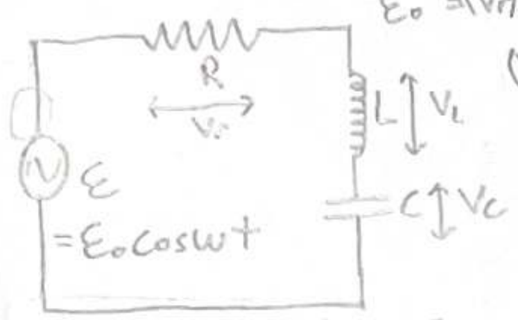


$\omega > \omega_c$; Transmits

$\omega < \omega_c$; Blocks

Lets high frequencies pass

Series RLC Circuit



$\mathcal{E}_0^2 = (V_r)_{max}^2 + [(V_L)_{max} - (V_c)_{max}]^2$ * if opposite, make ϕ negative

$(V_L)_{max} > (V_c)_{max}$ (assumed by diagram)

instantaneous Current: ϕ = Phase angle

$i = I_{max}\cos(\omega t - \phi)$

Peak Current: $I_{max} = \frac{\mathcal{E}_0}{\sqrt{R^2 + (X_L - X_c)^2}}$

Peak Voltages:

$(V_r)_{max} = (I_{max})(R)$

$(V_L)_{max} = (I_{max})(X_L)$

$(V_c)_{max} = (I_{max})(X_c)$

$\mathcal{E}_0^2 = [R^2 + (X_L - X_c)^2]I_{max}^2$

Impedance

"measured in Ω's"

$Z = \sqrt{R^2 + (X_L - X_c)^2}$

so... $I_{max} = \frac{\mathcal{E}_0}{Z}$

Phase Angle

$(V_r)_{max} = \mathcal{E}_0\cos\phi$

$\phi = \tan^{-1}\left(\frac{X_L - X_c}{R}\right)$

$\phi < 0$: • Current Leads • $\omega < \omega_0$

$\phi = 0$: • In Phase • $\omega = \omega_0$

$\phi > 0$: • $\mathcal{E}$ leads • $\omega > \omega_0$

Resonance

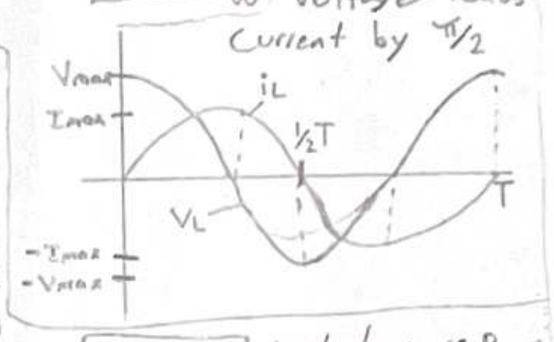
"where current is maxed"

occurs when: $X_L = X_c$

$\omega_0 = \frac{1}{\sqrt{LC}}$ (resonance frequency)

$Z = R$ at resonance

so $I_{max} = \frac{\mathcal{E}_0}{R}$



Power

instantaneous Power: $p = i\mathcal{E}$

Resistor instantaneous: $p_r = (I_r)_{max}^2 R\cos^2\omega t$

avg power dissipated by Resistor: $(P_r)_{avg} = (I_{rms})^2 R = \frac{(V_{rms})^2}{R} = I_{rms}V_{rms}$

Capacitor instantaneous lost: $p_c = -\frac{1}{2}\omega C(V_c)_{max}^2\sin(2\omega t)$ "Average power dissipate for P_L and $P_c = 0$" "positive when giving back power"

avg power supplied by EMF: $(P_{source})_{avg} = (I_{rms})(\mathcal{E}_{rms})$ [R circuit]

Average Power Supplied by EMF

$(P_{source})_{avg} = \frac{1}{2}(I_{max})\mathcal{E}_0\cos\phi = (I_{rms})(\mathcal{E}_{rms})\cos\phi$

where $\cos\phi$ is the power factor

Maxwell's equations in the Vacuum:

$\vec{E}$ Guass $\oint \vec{E}\cdot d\vec{A} = \frac{q}{\epsilon_0}$ | B Guass $\oint \vec{B}\cdot d\vec{A} = 0$

Faraday $\oint \vec{E}\cdot d\vec{r} = -\frac{d\Phi_B}{dt}$ | Ampere $\oint \vec{B}\cdot d\vec{r} = \cancel{\mu_0 I}^{0} + \mu_0\epsilon_0\frac{d\Phi_E}{dt}$

31 Electromagnetic Waves

Wave equation

$\frac{d^2E}{dx^2} = \epsilon_0\mu_0\frac{d^2E}{dt^2}$

$\frac{d^2B}{dx^2} = \epsilon_0\mu_0\frac{d^2B}{dt^2}$

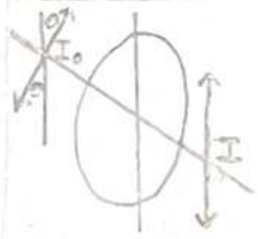
Wave speed

$v = \frac{\omega}{k}$ $E = cB$

$v = \frac{1}{\sqrt{\epsilon_0\mu_0}} = f\lambda = c$ $k = 2\pi/\lambda$

Polarization

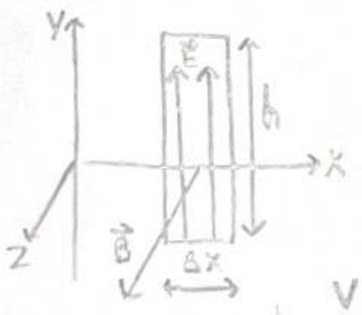
Law of Malus:



$I = I_0\cos^2\theta$

where I is the intensity of the wave. θ is the angle between the field and the polarizer axis

Faraday's Law (x-y) Plane



$\frac{d\Phi_B}{dt} = h\Delta x\frac{dB}{dt}$

$\frac{dE}{dx} = -\frac{dB}{dt}$

Connecting E and B

$E_y(x,t) = E_0\sin(kx - \omega t)\hat{j}$

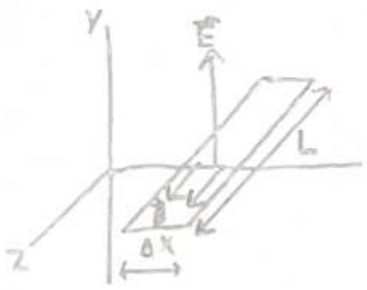
$= E_0\sin\left[2\pi\left(\frac{x}{\lambda} - ft\right)\right]\hat{j}$

$B_z(x,t) = B_0\sin(kx - \omega t)\hat{k}$

$= B_0\sin\left[2\pi\left(\frac{x}{\lambda} - ft\right)\right]\hat{k}$

Note $E_0 = \lambda f B_0$

Ampere's Law (x-z) plane



$\frac{d\Phi_E}{dt} = L\Delta x\frac{dE}{dt}$

$\frac{dB}{dx} = -\epsilon_0\mu_0\frac{dE}{dt}$

Energy and Intensity

Poynting Vector ($\vec{S}$): $\vec{S} = \frac{1}{\mu_0}\vec{E}\times\vec{B}$

$|S| = c\epsilon_0 E^2$

$S_{max} = \frac{E_0^2}{c\mu_0}$

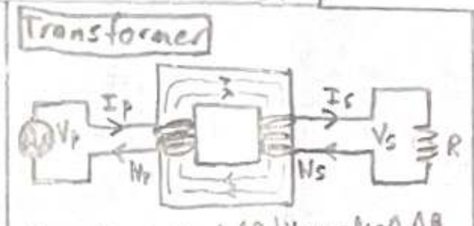
$I = \frac{Power}{Area}$

Intensity (I) $= S_{avg} = \frac{1}{2c\mu_0}E_0^2 = \frac{c\epsilon_0}{2}E_0^2 = \frac{E_0^2}{2\mu_0 c}$

radiation pressure: $P_{rad} = \frac{P/A}{c} = \frac{I}{c}$

radiation force: $F = \frac{Power}{c} = \frac{U/\Delta t}{c} = \frac{\Delta p}{c}$ momentum

Transformer



$V_p = \mathcal{E} = -N_p A\frac{\Delta B}{\Delta t}$ | $V_s = -N_s A\frac{\Delta B}{\Delta t}$

$\frac{V_s}{V_p} = \frac{N_s}{N_p}$ Power: $P_p = V_pI_p = V_sI_s = P_s$

$V_s = \frac{N_s}{N_p}V_p$ $N_s > N_p$; Step up. $N_p > N_s$; Step down

Angular

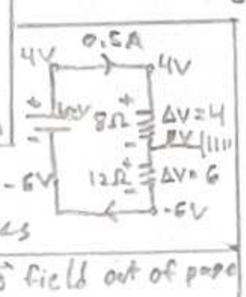
$\omega = \frac{2\pi}{T} = 2\pi f$

$f = 1/T$; $T \sim \lambda$

$f = c/\lambda$; $\lambda = c/f$

Ground

Changes nothing except specific voltages



Hall effect

$\vec{B}$ field out of page

$\Delta V_H = \frac{IB}{tne}$

Ring Spin

$\vec{B} = \frac{\mu_0}{4\pi}\frac{QR^2\omega}{(R^2+x^2)^{3/2}}$

Rotating Coil in a $\vec{B}$ field

$\vec{B} = \frac{\mu_0\omega Q}{2\pi R^2}\left[\sqrt{x^2+R^2} + \frac{x^2}{\sqrt{x^2+R^2}} - 2x\right]$

Units:

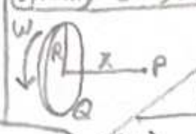
Current – A
Charge – C
$\mathcal{E}/\Delta V$ – V
R/X/L/Z – Ω
Power – W
Capacitance – F
Φ_E – V·m

Inductance – H
Φ_B – Wb
E field – V/m
B field – T

$qvB = \frac{mv^2}{r}$

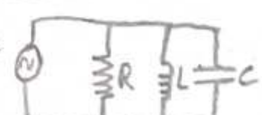
Key:

Spinning charged disk



Parallel RLC Circuit

$X_L = \omega L$ $X_c = \frac{1}{\omega C}$



$I_r = \frac{V}{R}$

$I_L = \frac{V}{X_L}$

$I_c = \frac{V}{X_c}$

$I_s = \sqrt{I_r^2 + (I_L - I_c)^2}$

$Z = \sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{X_c} - \frac{1}{X_L}\right)^2}^{-1}$

$(V_r)_{rms} = I_{rms}R$

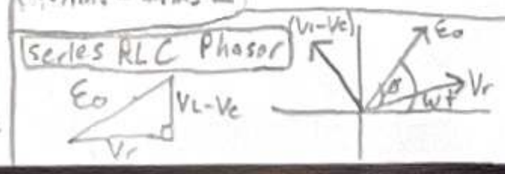
$(V_c)_{rms} = I_{rms}X_c$

$(V_L)_{rms} = I_{rms}X_L$

$(V_s)_{rms} = I_{rms}Z$

$\mathcal{E} = NBA\omega\sin\omega t$

Series RLC Phasor



Flip Coil

Flux before: NB·A

Flux after: −NB·A

$\Phi_B = -2NBA$

$\Delta Q = \frac{2NBA}{R}$ $B = \frac{VRC}{2NA}$

Rail Gun

(Motional EMF) "Work done by pulling or pushing exactly matches Resistor dissipate"

Power $= FV = \frac{V^2L^2B^2}{R}$

Amperes Law

Given:
- ∞ thin wire with Current = I
- Thick shell, uniform Charge density with Current = −2I

Find: $B(r) = ?$

$\oint_{L_1}\vec{B}\cdot d\vec{l} = \mu_0 I$

$\oint B\,dl\cancel{\cos\theta}^{1} = \mu_0 I$

$B2\pi r = \mu_0 I$

$B = \frac{\mu_0 I}{2\pi r}$; $r < a$

$\oint_{L_1}\vec{B}\cdot d\vec{l} = \mu_0(-2I + I)$

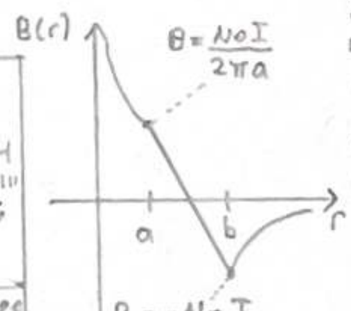
$\oint B\,dl\cancel{\cos\theta}^{1} = -\mu_0 I$

$B2\pi r = -\mu_0 I$

$B = -\frac{\mu_0 I}{2\pi r}$; $r > b$

$\oint_{L_2}\vec{B}\cdot d\vec{L} = \mu_0\left(I - 2I\cdot\frac{A_2}{A_{tot}}\right)$

$\oint B\,dl\cos\theta = \mu_0\left(I - 2I\frac{\pi(r^2 - a^2)}{\pi(b^2 - a^2)}\right)$

$B2\pi r = \mu_0 I\left(1 - \frac{2(r^2 - a^2)}{b^2 - a^2}\right)$

$B = \frac{\mu_0 I}{2\pi r}\left(1 - \frac{2(r^2 - a^2)}{b^2 - a^2}\right)$; $a < r < b$